

How the Light Controller Works

Thursday, July 9, 2026 6:23 PM

Light Control Circuit Operation

The control board is powered on, and its LED indicator is illuminated. The wall-mounted light switch is also in the **ON** position, and the ceiling LED lights are on.

The electromechanical switch on the control board is initially in the **normally open (NO)** position and is waiting for a command from the handheld IR controller.

Turning the Lights Off with the IR Controller

I point the handheld IR controller at the IR receiver and press the button for **Zone 1**.

The control circuit responds by turning off the LED indicator on the control board and switching the ceiling lights off. The wall-mounted light switch remains in the **ON** position.

A **15-second debounce timer** then begins. During this period, the circuit waits for the debounce cycle to complete.

After the 15 seconds have elapsed, the ceiling lights remain off, the control-board LED remains off, and the wall switch is still in the **ON** position.

Turning the Lights Back On

After the 15-second debounce period has completed, I press the **Zone 1** button on the IR controller again.

The ceiling lights turn back on, and the LED indicator on the control board also turns back on. The wall-mounted light switch remains in the **ON** position.

The circuit is now operating normally and is ready to receive another IR command.

Resetting the Circuit with the Wall Switch

If I press the **Zone 1** button on the IR controller again, the control-board LED turns off and the ceiling lights turn off. The 15-second debounce timer begins again.

After the 15-second debounce period has completed, I can now turn the wall-mounted light switch to the **OFF** position.

Turning the wall switch **OFF** causes the LED indicator on the control board to turn back **ON**, which resets the control circuit.

I can then turn the wall-mounted switch back to the **ON** position. The ceiling lights come back on, and the control circuit is once again ready to receive a command from the IR controller.

At this point, pressing the **Zone 1** button on the IR controller will turn the ceiling lights off again, and the

same sequence repeats.

Normal Operating Cycle

In summary, the circuit operates as follows:

Wall switch **ON** → Ceiling lights **ON** → IR command → Lights **OFF** → 15-second debounce →

Then:

IR command → Lights **ON** →

Then:

IR command → Lights **OFF** → 15-second debounce →

Then:

Wall switch **OFF** → Circuit resets → Wall switch **ON** → Lights **ON** → Ready for next IR command.